

Appendix A: Session-Based Testing (SBT) Report

EchoClass - Real-Time Student Engagement Platform

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Environment: macOS, Google Chrome (with DevTools mobile emulation), and Mozilla Firefox

A.1. Testing Methodology

On this test, the Session-Based Test Management (SBTM) was used as the exploratory testing methodology to test the system of EchoClass development [\[1\]](#). SBTM uses time-boxed sessions, each governed by a charter that defines the scope and objectives. This method was chosen because automated tests only verify code logic, hence they cannot evaluate whether the interface is usable on a mobile phone or whether identity switching works properly in a real-time session [\[2\]](#). Each session followed the PROOF framework (Past, Results, Obstacles, Outlook, and Feelings) for the debrief process. Testing was performed on macOS using Google Chrome with DevTools mobile emulation (iPhone SE 375x667, iPhone 12 390x844) and Mozilla Firefox for cross-browser verification. Two browser windows were used simultaneously: one logged in as a student and one as a lecturer, as well as in private browsing mode and on a real mobile device, enabling real-time verification of cross-role interactions.

Six sessions were conducted covering the three research questions (RQ1: identity modes, RQ2: lecturer analytics, RQ3: student self-reflection), non-functional security requirements, and overall mobile user experience. The total testing time was approximately 4 hours. The hybrid approach of combining the exploratory testing with the automated test suites follows the model that is established by Shah et al. [\[3\]](#), here explained that unifying exploratory and scripted testing provided broader defect coverage than just either method alone.

A.2. Summary of Results

Session	Charter	RQ	Duration	Bugs	UX	Severity	Action
1	Mobile and Desktop User Experience	All	60 min	10	3	Medium	Fixed
2	Analytics Dashboard (Lecture)	RQ2	45 min	8	2	Medium	Fixed
3	Student Self-Reflection	RQ3	45 min	7	3	Medium	Fixed
4	Identity Mode Switching	RQ1	35 min	6	0	High	Fixed
5	Security Verification	NFR	35 min	4	0	Medium	Fixed
6	AI Summary and Keyword Quality	RQ2	40 min	3	0	Low	Fixed

From all the six sessions, 38 defects were identified. All issues were resolved and all critical defects included a privacy-violating reply-to display that leaked anonymous users' real names and a missing Chart.js dependency that crashed the semester trend chart.

A.3. Detailed Session Reports

Session 1: Mobile and Desktop User Experience

Research Question: All RQs

Charter: Explore the student chat interface on a mobile device (Chrome DevTools mobile emulation). Can a student join, send messages, read messages, switch identity modes, reply, react, and vote in polls without frustration on a 375px screen?

Duration: 60 min | Bugs Found: 10 | UX Issues: 3 | Overall Severity: Medium

Findings:

ID	Finding	Severity	Action Taken
1.1	Double-escaping of apostrophes in messages: server <code>sanitiseText()</code> used <code>validator.escape()</code> converting apostrophes to HTML entities, this makes it display as a raw <code>&#x27;</code> to users.	Medium	Fixed: replaced <code>validator.escape()</code> with <code>validator.stripLow()</code> in <code>security.js</code> . This makes the client-side <code>escapeHtml()</code> now serves as the single XSS defence layer.
1.2	Deleted announcement messages persisted in the announcement banner.	Medium	Fixed: added announcements array filtering and call to the message-deleted handler in <code>student-chat.js</code> .
1.3	Pin emoji rendering as Unicode code point <code>U0001F4CC</code> -CSS content property used incorrect escape syntax (<code>\U0001F4CC</code> instead of the actual emoji character).	Low	Fixed: replaced CSS escape with the correct emoji character in <code>messages.css</code> .
1.4	Soft-deleted messages still loading from database: <code>GET /session/:sessionId</code> query did not filter <code>isDeleted</code> field, returning deleted messages to client on page refresh.	Medium	Fixed: added <code>isDeleted: {\$ne: true}</code> filter to the <code>Message.find()</code> query in <code>messages.js</code> .
1.5	Redundant 'None' message type in dropdown rendered identically to Comment which this makes user confusion.	Low	Fixed: removed None option from <code>identity-mode.js</code> ; Comment become the default selection.

1.6	Lecturer could pin messages but cannot unpin.	Medium	Fixed: added updatePinUI() function to student-chat.js to pin and unpin.
1.7	Multi-select polls broken, click handler disabled after first vote.	Medium	Fixed: updated click gate logic and votePoll() to compute toggled option sets for multi-select mode.
1.8	Cannot send attachment without have a comment.	Medium	Fixed: sendMessage() now accepts text pendingAttachment. Server validation already accepted attachment-only messages.
1.9	JPG images are not displaying in room chat instead need to open the file with Preview/Download row appeared alongside the image.	Low	Fixed: stripped redundant row for image attachments.
1.10	Empty message body bubble rendered when sending attachment-only messages, showing unnecessary whitespace.	Low	Fixed: skip rendering. message-body entirely when text is empty and attachment is present.

What Worked Well:

From the UX side, the web login, identity switching mode, profile session, lecture and student reflection until the chatroom fits well on both desktop and mobile. The message colouring, padding, emojis used, etc shows perfectly. Then, the reply threading renders on narrow screens and every time scrolling through the chat is smooth. The orientation also adaptable between portrait and landscape view, and attachment upload and inline image (jpg, etc) display function correctly. Session 1 (Mobile UX) had the highest defect concentration at 10 defects, reflecting the complexity of the chat interface as the primary user-facing component.

Session 2: Analytics Dashboard (Lecture)

Research Question: RQ2

Charter: Explore the lecturer micro-analytics dashboard and macro semester analytics. Does each component display correct data and the visualisations readable and useful, included the AI summary produce relevant content, extended to the CSV export work and whether the cross-session trends render accurately?

Duration: 45 min | Bugs Found: 8 | UX Issues: 2 | Overall Severity: Medium

Findings:

ID	Finding	Severity	Action Taken
2.1	Stat card labels truncated with ellipsis on smaller viewports - 'Contributors / Total' and 'Messages/Minute' labels clipped.	Low	Fixed: shortened labels and removed white space: nowrap CSS rule.
2.2	--text-color CSS variable undefined in light mode — rules using var(--text-color) had no concrete light-mode value, falling back to inheritance.	Low	Fixed: added --text-color: #1e293b and --card-bg: #ffffff to: root in base.css.
2.3	Analytics modal inline styles prevented responsive CSS rules from applying pre-existing mobile rules in responsive.css were dead code because HTML lacked matching class names.	Medium	Fixed: replaced inline styles with semantic classes (.analytics-modal-content, .analytics-header, .analytics-header-buttons) wiring the responsive rules.
2.4	AI summary header displayed '19 students' instead of '19 contributors', wording in both analytics.js overview banner and ai-summary.js Hugging Face prompt template.	Low	Fixed: changed to 'contributors' in analytics.js:461 and to 'contributors' in ai-summary.js:48.
2.5	Participation rate calculation used wrong denominator when session spanned a long-time window.	Medium	Fixed: rate calc now uses coreMessages.length instead of totalMessages in analytics.js.
2.6	'Quietest period' wording in engagement pattern which this is unclear language.	Low	Fixed: changed to 'lowest-activity window' in analytics.js:469.
2.7	Demo data not topic-aware — Data Structures & Algorithms session contained messages about TCP/IP, DNS, and firewalls instead of DSA	Medium	Fixed: added TOPIC_MSGS map to demo-data.js with 8 topic-specific pools (web, db, oop, testing, hci, cloud, dsa, ml). 80% of messages now drawn from topic pool.

	content, making AI summaries incoherent.		
2.8	Analytics modal had no close mechanism other than the Close button. There is no backdrop click or escape key support.	Medium	Fixed: added backdrop click and Escape key event listeners in lecturer-dashboard.js.

What Worked Well:

On this session, firstly, from the 10 micro-analytics the components that are rendered correctly are the summary statistics card (total messages, the contributors, rate participation, and the message sent per minute), the engagement timeline correctly detected peak activity and the chart shows it correctly with the identity mode breakdown bar chart also give the correct amount on what is on the real room chat with the confusion, question, rate indicators. Here we can see that the system also already has the top keywords extracted the one related/relevant with the session topic with top contributors shown. The macro analytics also checked and evaluated the cross-session comparison table across six sessions, module filter dropdown, participation and confusion trend charts, identity trends stacked bar chart showing anonymous/pseudonymous/identified proportions per session, and recurring confusion hotspot detection identifying topics flagged across multiple sessions.

Session 3: Student Self-Reflection

Research Question: RQ3

Charter: Explore the student reflection features. Can a student set goals, view their stats, see badges, get tips, and submit reflections and check it with SRL theory (Forethought, Performance, Self-Reflection phases)?

Duration: 45 min | Bugs Found: 7 | UX Issues: 3 | Overall Severity: Medium

Findings:

ID	Finding	Severity	Action Taken
3.1	Semester trend chart crashed with 'Chart is not defined' error -Chart.js CDN script was only included in lecturer-dashboard.html, not in chat-room.html where the student reflection dashboard renders.	Critical	Fixed: added Chart.js CDN script to chat-room.html.

3.2	Session history change arrows displayed backwards -comparison used the wrong index direction, showing an increase when engagement had declined.	Medium	Fixed: changed comparison from prevMessages to history[i+1] in student-reflection.js.
3.3	Timeline insight displayed confusing 'Peak at 0min' text instead of a readable time range.	Low	Fixed: changed to '0-5min' format in getTimelineInsight().
3.4	Engagement timeline bars barely visible -fixed pixel scaling produced uniform tiny bars regardless of message count.	Medium	Fixed: implemented relative scaling: $(\text{count} / \text{maxCount}) * 100$ with 8px minimum height.
3.5	Achievement badges only appeared when earned: Above Average, Goal Achiever, Early Bird, and Top 10% badges were invisible when locked, preventing students from seeing what they could work towards.	Medium	Fixed: all six badges now always visible with earned (gold) and locked (greyed with hint text) states.
3.6	No badge tooltips: hovering over badges provided no explanation of criteria or how to earn them.	Low	Fixed: added title attributes showing badge name and unlock criteria.
3.7	Sessions with zero messages displayed 'Below' average instead of a neutral indicator.	Low	Fixed: zero-message sessions now show dash instead of misleading comparison.

What Worked Well:

Session 3 confirmed the goal setting and everything based on the SRL Forethought Phase which give a good usability to user (student). This includes showing the personal statistics grid that shows the message and contribution count, class average, rank, and the percentile of the student on each session with message type breakdown that the pie chart renders correctly. The goal progress bar updates also after they send the messages question (shows real-time). The reflection for submission saves the understanding rating, the confusion topic, and the improvement area there and the session history table displays past sessions with engagement data too.

Session 4: Identity Mode Switching

Research Question: RQ1

Charter: Systematically test all three identity modes by sending messages under each mode which here verify the correct display to other users, correct recording in analytics, and that anonymity guarantees are not violated through reply to contexts or edge cases.

Duration: 35 min | Bugs Found: 6 | UX Issues: 0 | Overall Severity: High

Findings:

ID	Finding	Severity	Action Taken
4.1	Reply-to display leaked real name of anonymous and pseudonymous senders -replying to an anonymous message showed the original sender's real display name in the reply to context, de-anonymising them. This is a privacy violation that directly undermines the anonymity guarantee of RQ1.	Critical	Fixed: replyTo populate now includes identityMode and alias fields; display name respects identity mode (anonymous shows 'Anonymous', pseudonymous shows alias, identified shows real name).
4.2	Cannot reply to users with apostrophes in names (O'Brien, O'Neill): escapeHtml() did not escape single quotes, breaking the onclick JavaScript string attribute.	Medium	Fixed: added escapeForAttr() helper that escapes backslash and single quotes for onclick attributes.
4.3	Lecturer messages stored as identityMode: 'anonymous': lecturer send path did not include identityMode field, and the model defaulted as 'anonymous'.	Medium	Fixed: lecturer send path now explicitly sets identityMode: 'identified'.
4.4	Dashboard analytics included lecturer messages in student-only metrics: lecturer COMMENT messages inflated the comment count and anonymous count in the message type and identity mode breakdowns.	Medium	Fixed: added userId: {\$ne: lecturerId} filter to totalMessages count, messagesByType aggregation, and identityModeBreakdown aggregation. CSV export also updated to filter student-only for summary statistics.

4.5	CSV summary statistics included lecturer messages also the type counts did not match dashboard figures.	Medium	Fixed: CSV filters to student-only for summary stats. Message Log still includes all messages with Role column.
4.6	CSV did not show deleted/reported message status.	Low	Fixed: Status column now shows DELETED, EDITED, REPORTED, or combinations (included in summary).

What Worked Well:

By testing the anonymity, the focus was on verifying that anonymous mode functions correctly with no real profile identification and tested between three of them. Each message switching even in fast-paced type and sending (there is limit for every minute regarding the security matter), and it shows exactly as what the last set user set. Macro analytics identity trends across sessions display correctly. After applying the reply-to fix, all three modes displayed correctly in both the chat and the analytics dashboard.

Session 5: Security Verification

Research Question: NFR

Charter: Attempt common security attacks to verify the protection layers.

Duration: 35 min | Bugs Found: 4 | UX Issues: 0 | Overall Severity: Medium

Findings:

ID	Finding	Severity	Action Taken
5.1	XSS messages displayed as escaped HTML entities (<script>) rather than readable plain text -the double-escaping bug from Session 1 also affected security test messages.	Medium	Fixed: resolved by the Session 1 sanitisation fix (validator.stripLow replaces validator.escape).
5.2	Rate limiter never triggered during manual testing: authLimiter was configured with max: 50 requests per 15-minute window, far too high for the 15-request curl test to hit.	Medium	Fixed: lowered to max: 10 per 15-minute window, providing effective brute-force protection while accommodating normal usage patterns.
5.3	Messages vanish on reload.	Medium	Fixed: changed sort to timestamp field which every message has.

5.4	Deleted messages fully disappeared on reload instead of showing the identification as deleted message.	Medium	Fixed: changed query to return all messages; client renders 'This message was deleted' placeholder both on initial load and real-time deletion.
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What Worked Well:

Security was verified through XSS injection that blocked across all three test vectors (script tag, img onerror, javascript: href) and all rendered as plain text with no script execution. The access control verified that the student accessing lecturer-only analytics is forbidden and the unauthenticated request to /api/sessions/my-sessions also forbidden which return the 401 Unauthorized. Then the security headers confirmed present via DevTools Network tab: X-Frame-Options (SAMEORIGIN), X-Content-Type-Options (nosniff), Referrer-Policy (strict-origin-when-cross-origin), X-XSS-Protection (1; mode=block). Lastly, X-Powered-By header confirmed absent. Rate limiter triggers correctly at 10 attempts, returning 429 Too Many Requests.

Session 6: AI Summary and Keyword Quality

Research Question: RQ2

Charter: Evaluate the quality and usefulness of the AI-generated session summary and keyword extraction. This covers testing with topic-aware demo data, verify fallback mode (RAKE) works when Hugging Face API is unavailable, and compare keyword accuracy against the ai-experiment.js benchmark results.

Duration: 40 min | Bugs Found: 3 | UX Issues: 0 | Overall Severity: Low

Findings:

ID	Finding	Severity	Action Taken
6.1	AI summary extracted lecturer administrative messages ('lab moved to room 01/009') as discussion content instead of student topic discussion.	Low	Fixed: filtered out lecturer-role messages from the DistilBART input; only student messages are fed to the summarisation model.
6.2	Duplicate questions appeared in the Key Questions section of the AI summary.	Low	Fixed: added text-based deduplication before slicing the top five questions.

6.3	RAKE fallback extracted filler words ('lost', 'finally', 'big', 'considered', 'clicks') and lecturer administrative terms ('peer review', 'group project', 'canvas', 'deadline') as keywords.	Low	Fixed: expanded the fallbackStopWords set with approximately 30 additional common filler and administrative terms.
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What Worked Well:

This AI session summary was generated successfully, this also including all six sections. The first is using DistilBART, which accurately identified the session topic in Data Structures and Algorithms as quicksort, recursion, Big O notation, and dynamic programming. Fallback mode was also confirmed. After commenting out HUGGINGFACE_API_KEY and restarting the server, the system produced a structured statistical report without the AI narrative paragraph, as seen in the UX itself, which summarises that using only the fallback (RAKE) is sufficient. The engagement pattern, message classification, and discussion topics sections remained available through RAKE keyword extraction, demonstrating graceful degradation. A comparison of AI experiment scripts showed Hugging Face achieved an average accuracy of 66.7%, while RAKE averaged 61.7%. The low overlap between methods (0-11%) indicates fundamentally different extraction strategies, supporting the value of a dual-method approach.

A.4. Impact Analysis (Changes Driven by Testing)

Below are the key changes made to the codebase as a direct result of SBT findings:

Finding	Severity	Impact on System	Change Implemented
4.1: Reply leaks real name	Critical	Directly undermined RQ1 anonymity guarantee. Users who believed they were anonymous could be identified through the reply to context.	Refactored replyTo population to include identityMode and alias; display logic now respects the original sender's chosen identity mode across all three rendering paths (HTTP load, Socket.IO broadcast, reply to context).
3.1: Chart.js missing	Critical	Semester trend chart (RQ3 Self-Reflection Phase) was completely non-functional for students.	Added Chart.js CDN dependency to chat-room.html. Semester trend and session history charts now render correctly.

1.1: Double escaping	Medium	All messages containing apostrophes displayed corrupted text, affecting readability across the entire chat interface.	Replaced <code>validator.escape()</code> with <code>validator.stripLow()</code> as the server-side sanitisation strategy. Client-side <code>escapeHtml()</code> now serves as the single XSS defence layer, eliminating double-encoding.
4.3: Lecturer identity mode	Medium	Identity mode analytics (RQ1) were corrupted by lecturer messages being classified as anonymous, inflating anonymous counts by the number of lecturer messages per session.	Lecturers send path explicitly sets <code>identityMode: 'identified'</code> . Dashboard aggregations filter to student-only messages for identity analytics.
5.2: Rate limiter threshold	Medium	Authentication rate limiter configured at 50 requests per window provided insufficient brute-force protection.	Reduced to 10 requests per 15-minute window, aligning with OWASP recommendations for authentication endpoints.
2.7: Demo data mismatch	Medium	AI summaries produced incoherent content because session messages did not match session topics.	Added topic-aware message generation with 8 curated pools covering 80% of messages per session.
6.1: AI summary content	Low	DistilBART summarisation included lecturer administrative messages, producing summaries about room changes rather than student discussion.	Filtered lecturer-role messages from the summarisation input, improving topic relevance.
1.7: Multi- select polls	Medium	Students could not select multiple options in polls designed for multiple answers, breaking a core interaction feature.	Rewrote click gate and vote toggle logic to support true multi-select with option toggling.
5.3: Messages vanish on reload	Medium	Messages sent by real users disappeared on page refresh because the sort field (<code>createdAt</code>) was null for non-seeded messages.	Changed sort field from <code>createdAt</code> to timestamp which every message has.

3.5: Hidden unearned badges	Medium	Students could not see which badges existed or how to earn them, undermining the SRL motivational design.	All six badges now always visible with earned (gold) and locked (greyed with hint text) states.
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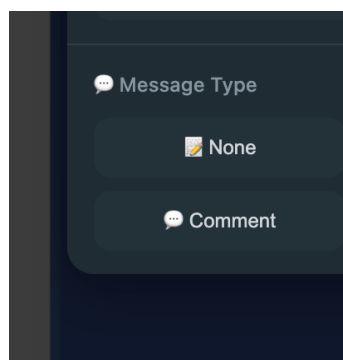
A.5. Conclusion

Session-Based Testing was effective, resulting in the identification of defects that automated testing missed. While all the automated Jest tests confirmed code correctness at the unit and integration levels, SBT sessions uncovered interaction-level issues and the real visual defects, including a critical privacy violation in reply-to display (Finding 4.1), a double-escaping rendering defect (Finding 1.1), and corrupted identity analytics from unlabelled lecturer messages (Finding 4.3). The highest defect concentration was in Session 1 (Mobile UX, 10 defects), since that is the chat interface and has the most complex client-side component, which required the most iterative refinement. These bugs only surfaced during manual cross-role testing, first opening the student chat in one browser and the lecturer dashboard in another, then checking whether anonymous messages stayed anonymous in the reply context between student and lecturer windows, switching identity modes during conversations, and visually inspecting rendered output. A total of 38 defects were found during formal SBT sessions: 2 critical, 15 medium, and 21 low severities. All issues were addressed before submission, and the testing directly led to code changes that improved the system's correctness, security, privacy, and usability.

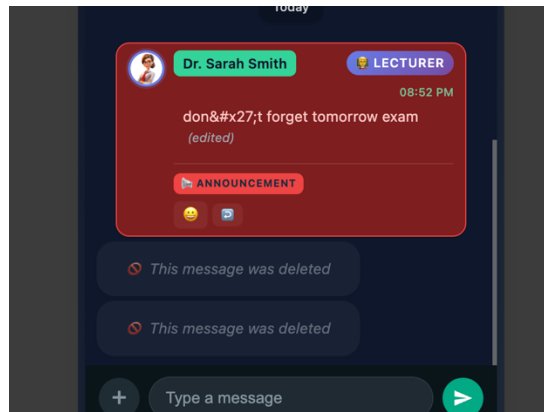
A.6. Bugs documentation

** The full bug documentation with screenshots is available in the project GitLab repository*

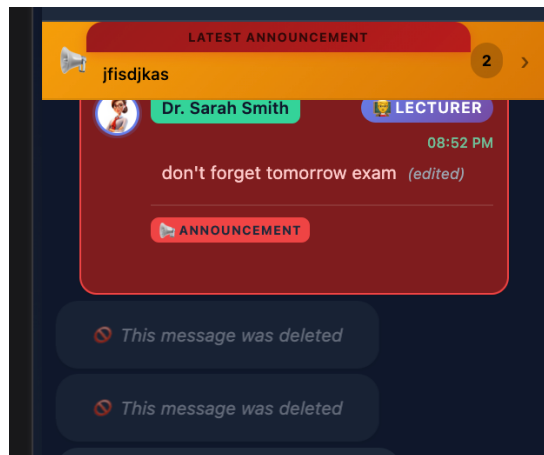
Removing None Type



Double-escaping apostrophes



Announcement not removed if message deleted

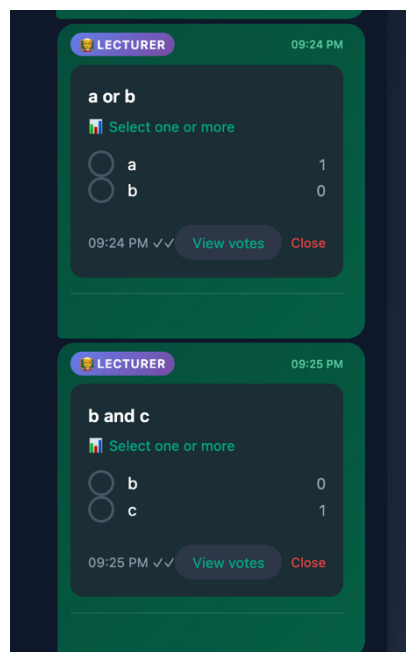


Multi-select vote does not work

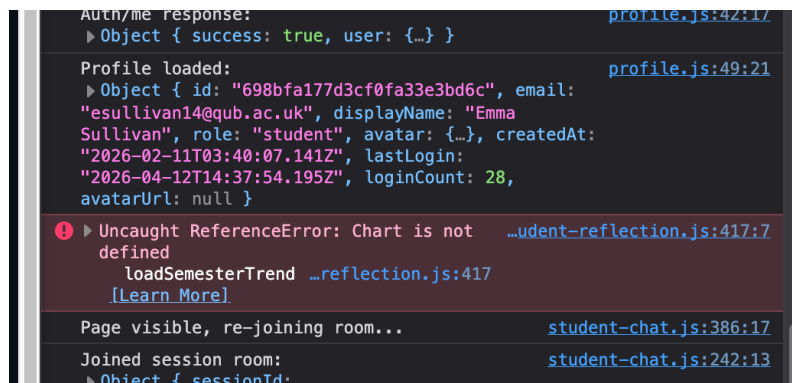
```

Initializing announcement & pin feature  announcement-pin.js:7:11
▶ XHR GET http://localhost:3000/api/me... [HTTP/1.1 404 Not Found 8ms]
! ▶ View votes error: SyntaxError: JSON.parse: student-chat.js:1315:29
unexpected character at line 1 column 1 of
the JSON data

```



Error Chart on reflection analytic



```
Auth/me Response: profile.js:42:17
  ▶ Object { success: true, user: {...} }

Profile loaded: profile.js:49:21
  ▶ Object { id: "698bfa177d3cf0fa33e3bd6c", email:
    "esullivan14@qub.ac.uk", displayName: "Emma
    Sullivan", role: "student", avatar: {...}, createdAt:
    "2026-02-11T03:40:07.141Z", lastLogin:
    "2026-04-12T14:37:54.195Z", loginCount: 28,
    avatarUrl: null }

❗ Uncaught ReferenceError: Chart is not
  defined student-reflection.js:417:7
    loadSemesterTrend ...reflection.js:417
    [Learn More]

Page visible, re-joining room... student-chat.js:386:17

Joined session room: student-chat.js:242:13
  ▶ Object { sessionId:
```

References

- [1] J. Bach, "Session-Based Test Management," Software Testing & Quality Engineering, vol. 2, no. 6, pp. 32–37, 2000.
- [2] J. Itkonen, M. V. Mäntylä, and C. Lassenius, "Defect Detection Efficiency: Test Case Based vs. Exploratory Testing," in Proc. 1st Int. Symp. Empirical Software Engineering and Measurement (ESEM), 2007, pp. 61–70. doi: 10.1109/ESEM.2007.56.
- [3] S. M. A. Shah, C. Gencel, U. S. Alvi, and K. Petersen, "Towards a hybrid testing process unifying exploratory testing and scripted testing," J. Software: Evolution and Process, vol. 26, no. 2, pp. 171–188, 2014. doi: 10.1002/smr.1610.